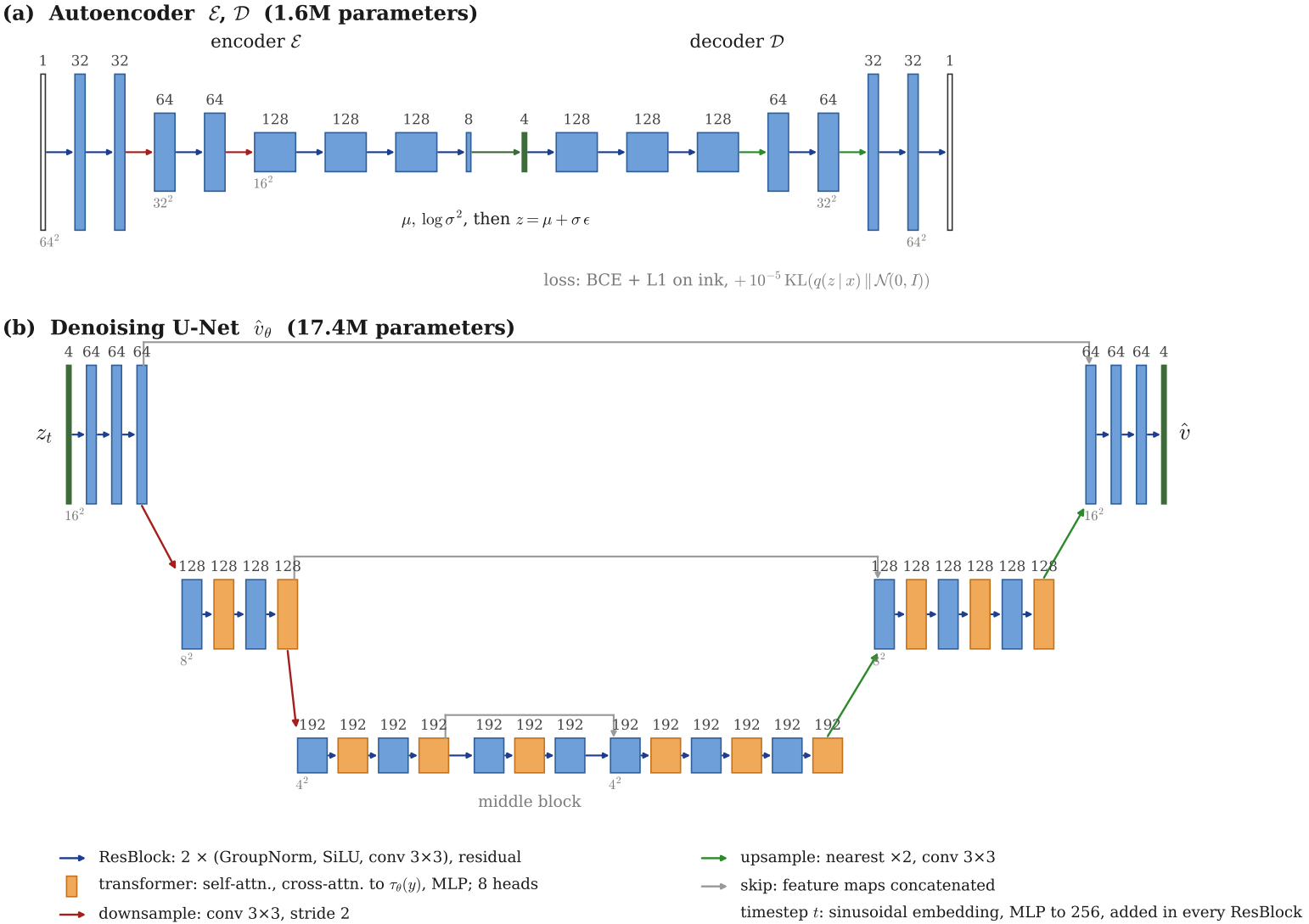

(a) Autoencoder $\mathcal{E}$, $\mathcal{D}$ (1.6M parameters)
encoder $\mathcal{E}$
decoder $\mathcal{D}$
1 32 32 64 64 128 128 128 8 4 128 128 128 64 64 32 32 1
64^2 32^2 16^2 32^2 64^2
μ, $\log\sigma^2$, then $z = \mu + \sigma\epsilon$
loss: BCE + L1 on ink, $+10^{-5}\,\mathrm{KL}(q(z\,|\,x)\,\|\,\mathcal{N}(0, I))$
(b) Denoising U-Net $\hat{v}_\theta$ (17.4M parameters)
z_t
4 64 64 64
64 64 64 4
$\hat{v}$
16^2
128 128 128 128
128 128 128 128 128 128
8^2
192 192 192 192 192 192 192 192 192 192 192 192
4^2
middle block
ResBlock: 2 × (GroupNorm, SiLU, conv 3×3), residual
transformer: self-attn., cross-attn. to $\tau_\theta(y)$, MLP; 8 heads
downsample: conv 3×3, stride 2
upsample: nearest ×2, conv 3×3
skip: feature maps concatenated
timestep t: sinusoidal embedding, MLP to 256, added in every ResBlock